🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

Write and Solve Equations Using Addition or Subtraction

Lesson 8-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve one-step addition and subtraction equations using inverse operations.

Language Objective: I can explain my steps using the words equation, inverse operation, isolate, and solution.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Equation | Inverse operation | Isolate | Solution | One-step equation | Balance

👆 *Tap any word to see its meaning and a picture.*

1. A math sentence with an equal sign showing two equal amounts is an ________.

2. An operation that undoes another, like subtraction undoing addition, is an ________.

3. To get the variable alone on one side of the equation is to ________ it.

4. The value of the variable that makes the equation true is the ________.

5. An equation solved in a single step is a ________.

6. Doing the same thing to both sides to keep them equal keeps the equation in ________.

Write About the Math

How far has the hiker walked? How did you use inverse operations?

¿Qué distancia ha caminado la excursionista? ¿Cómo usaste las operaciones inversas?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Equation** — *Ecuación*
- [ ] **Inverse operation** — *Operación inversa*
- [ ] **Isolate** — *Despejar*
- [ ] **Solution** — *Solución*
- [ ] **One-step equation** — *Ecuación de un paso*

Model: *I can undo the addition to get the variable by itself.* — Students recognize 23 is added to n and that subtracting 23 (the inverse operation) from both sides will isolate n.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The hiker walked ____ miles because d + 4.5 = 12 means d = 12 − ____ = ____.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Inverse operation
3. **Notes 3:** Isolate
4. **Notes 4:** Solution
5. **Notes 5:** One-step equation
6. **Notes 6:** Balance
7. **Watch for this mistake:** *A common mistake is applying the same operation instead of the inverse — for example, solving $x - 7 = 15$ by subtracting 7 from both sides (getting the wrong $x = 8$) instead of adding 7 to both sides (the correct $x = 22$). Another common mistake is doing the inverse operation to only one side, like solving $x + 6 = 14$ by subtracting 6 from the left but forgetting the right side. Before you submit, name the operation shown in the equation, choose its inverse, apply it to BOTH sides, and check by substituting your answer back into the original equation.*
8. **Try It 1:** A. $n = 13$ — *Add 4 to both sides: $n = 9 + 4 = 13$. Check: $13 - 4 = 9$ ✓*
9. **Try It 2:** A. 34 — *Subtract 16 from both sides: $t = 50 - 16 = 34$. Check: $34 + 16 = 50$ ✓*